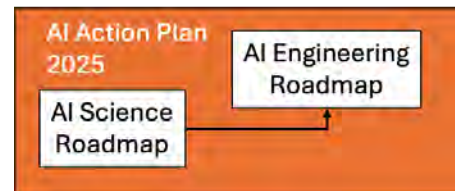


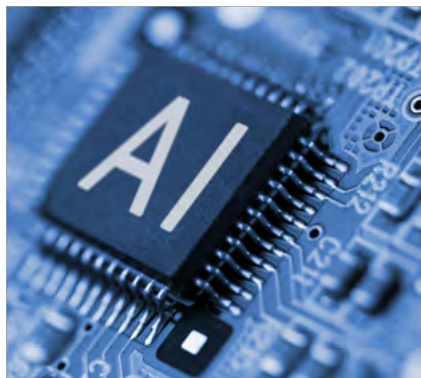
RFI for
Information on the Development of an
Artificial Intelligence (AI) Action Plan (3-1-2025)

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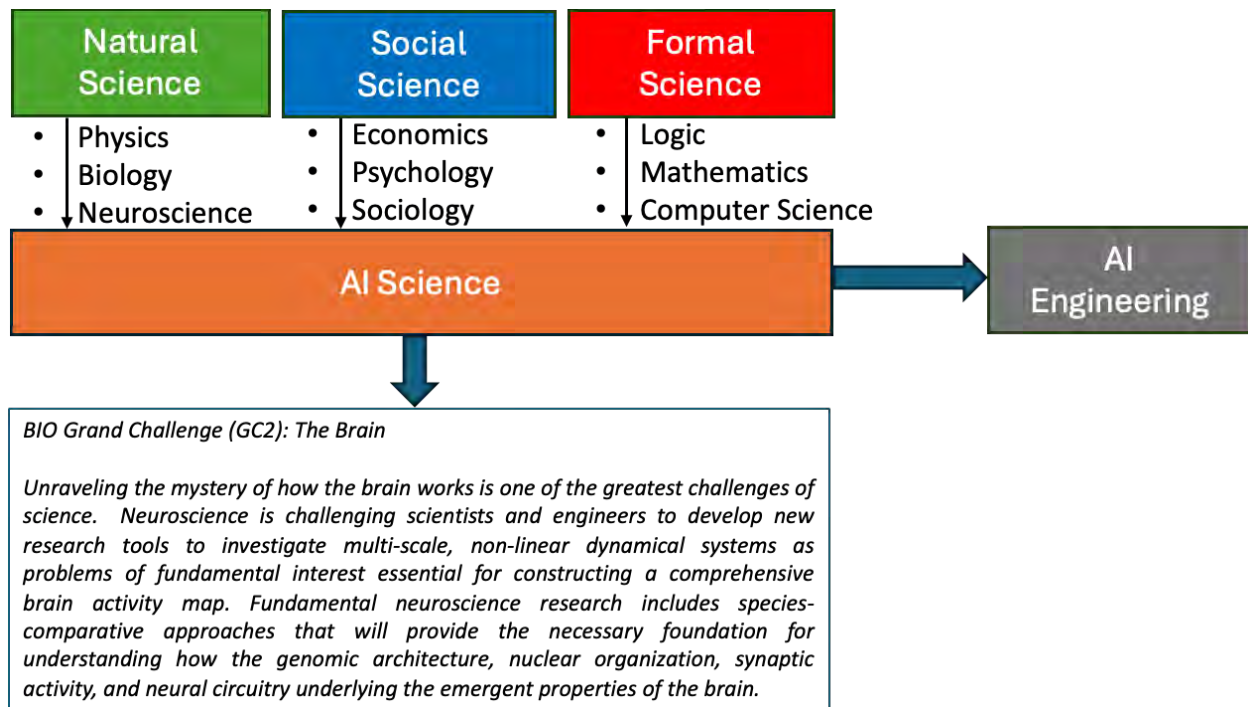
This Action Plan will define the priority policy actions needed to sustain and enhance America's AI dominance, and to ensure that unnecessarily burdensome requirements do not hamper private sector AI innovation. Artificial Intelligence (AI) has two distinct disciplines: science and engineering, and each discipline has a different purpose. The science of AI, or more specifically human intelligence, is focused on understanding the problem of how the brain and its intelligence works. In contrast, AI engineering also has a very specific purpose: to design a working AI system or “machine”. Engineering develops the solution. Both of these disciplines are important, but for different reasons. Research learned in AI science will have direct benefits to AI engineering, whose goal is to field a working prototype. This effort validates the maturity level of the technology and as a perspective, the ultimate robotic system is a humanoid.



Science is a systematic discipline that builds and organizes knowledge in the form of testable hypotheses and predictions about the universe. **AI Science** builds testable hypotheses and predictions about the human mind. Modern science is typically divided into two or three major branches that impact AI: the natural sciences (e.g., physics and biology), which study the physical world; and the social sciences (e.g., economics, psychology, and sociology), which study individuals and societies. The formal sciences (e.g., logic, mathematics, and computer sciences, which study formal systems governed by axioms and rules) are sometimes described as being science as well; however, they are often regarded as a separate field because they rely on deductive reasoning instead



of following the scientific method or empirical evidence as their main methodology. Thus, AI Science involves the natural sciences, social sciences, and to a major extent, the formal sciences. Biology includes neuroscience, the primary model for the neural network. Designing smart machines depend on physics, requiring economics that can support decision making (especially Von Neumann and Morgenstern's Expected Utility Function), requiring psychology through AI behaviors, and requiring sociology with human-machine interfaces and swarms. AI constitute all the formal sciences of logic, mathematics, and computer science.

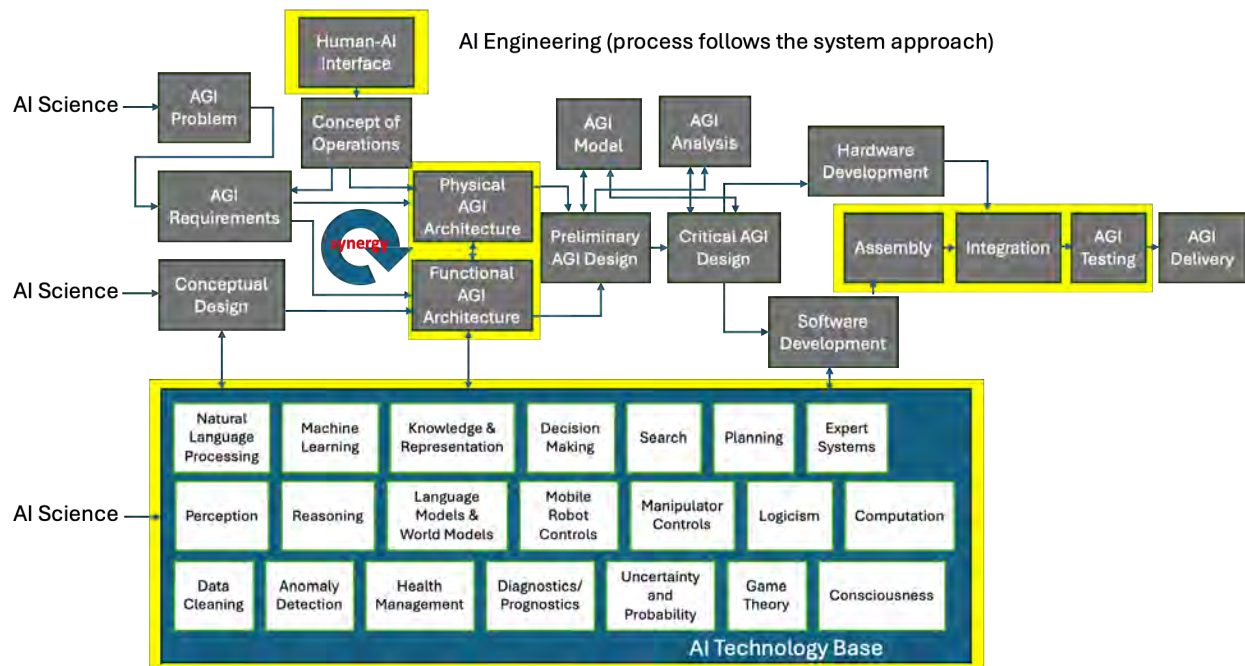


AI Science Roadmap

The main focus in AI science is answering the question of how does the human mind work? It has roots in cognitive neuroscience, philosophy, psychology, and logicism. By discovering the “science” to this arena, it will enable the engineering solution to emerge as a consequence of this work. Unfortunately, this is an incredibly hard problem with minimal progress advanced since monism, dualism, functionalism, determinism, etc. The main goal in AI Engineering is to achieve Artificial General Intelligence (AGI). This is also a hard problem but is ripe for innovation. AI engineering is a field populated by major AI companies such as Meta, Google, OpenAI, Amazon, Baidu, Tencent, etc. The products coming from these companies are considered engineering solutions. Now that we understand the breath and complexity of the problem, we can further develop an Action Plan. The approach taken here is based on being holistic and strategic. This design problem and its associated inter-connections are very broad. **Design** is the cornerstone of **engineering and** requires a structured approach (by using a systematic problem-solving strategy). **Systems engineering** is a set of steps (objects or elements) that interact to achieve a specific goal. In this case, the specific goal is AGI. **Systems Engineering** is a multi-disciplinary process and a means to delivering successful results. It is the basis for a holistic design and is used consistently in aerospace and the military and viewed as a well-established discipline for engineers. It also accounts for the “ilities” such as safety, redundancy, reliability, and maintenance. This approach embraces the system philosophy of *keeping it simple stupid* or KISS principle. The approach is essential. When we take a wider perspective of the problem, we find science has been the primary domain of academia while engineering has been the primary domain of

industry. The government ties everything together and it's important role is management, providing insight, oversight, and guidance.

Systems engineering is the applied aspects of system theory, a cohesive group of interrelated, interdependent components that can be natural (human intelligence) or artificial (artificial intelligence). The various AI components are captured here in something that we refer to as the AI Technology base (see next figure). Basically, every system has casual boundaries, is influenced by their context, is defined by its structure, function and role, and is expressed through its relations with other systems. **Engineering** is the practice of using natural science, mathematics, and the engineering design process to solve problems within technology, increasing efficiency and productivity, and improving designs. As such, **AI Engineering** is the application of AI Science to the optimum conversion of the resources of nature to the uses of humankind. Please pay special attention to the highlighted boxes. The process shown in the diagram above the AI Technology Base is the systems engineering process.

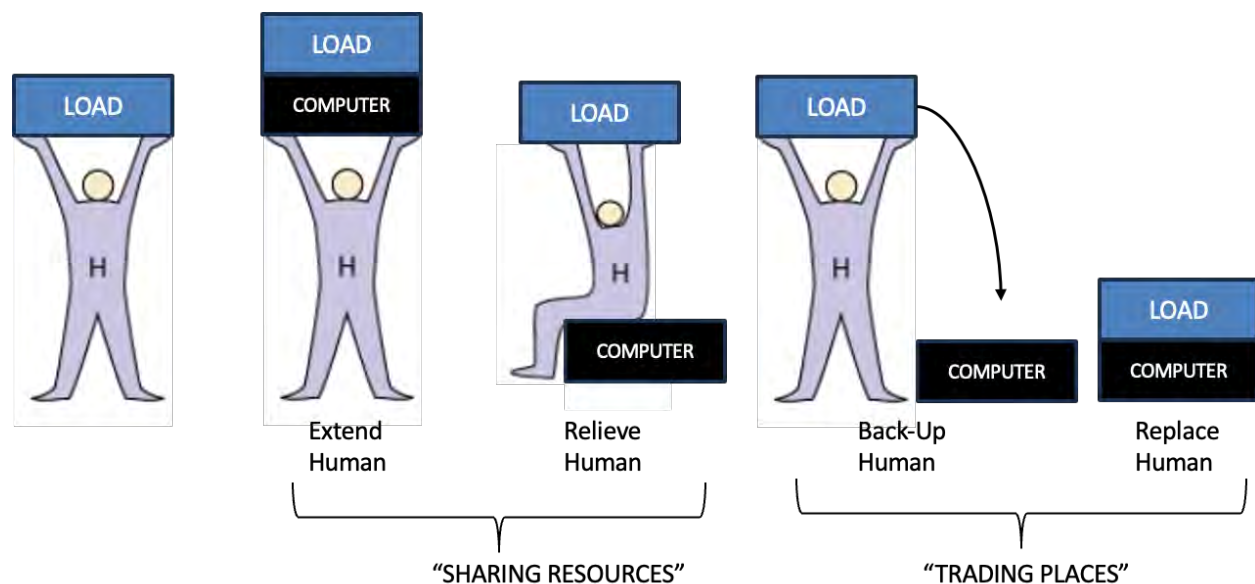


AI Engineering Roadmap

The past seventy years have produced a wide breath of AI technologies collected into something that we refer to here as an AI technology base. In recent years, this collection of AI technologies has been populated by deep learning, transformers, large language models, etc. These AI technologies may seem random, but popularity between connectionist theories and symbolic manipulation theories have switched and fluctuated over time to being in vogue at the time. It is our intention to individually evaluate the various technologies available which is integral to the system engineering process of

developing AGI. In typical large programs, management runs parallel to the design process. It is the job of management to assess each technology and to summarize each technology in terms of maturity (can also be carried forward as a risk analysis activity). NASA and DOE use the Technological Readiness Level (TRL), a maturity scale used to understand critical technologies. Unfortunately, the ratings can be subjective and may not have the required resolution necessary to accurately determine the technologies' readiness as required when needed. Past experience has shown that for any technology, there could be parts that are more mature than other parts within the same technology. More importantly, each rating should be justified with a rationale statement, similar to the requirements' rationale statement (which expands further justification) as defined in the system engineering process.

The AI Science roadmap is simple and straight-forward. This roadmap leaves room for research to continue to someday answer some critical issues about the human mind. And the majority of the AI Action Plan 2025 will focus on AI Engineering with the purpose of developing and fielding AI machine(s) by integrating hardware and software. Systems engineering follows an established process: starting with the problem, followed by developing the requirements, and next a step to come to consensus on a concept of operations (CONOPs). But prior to developing the concept of operations, a specific decision must be made on the human-computer interface, and whether the system is fully autonomous, automatic, or a shared situation between a human operator and its software.



Spectrum of Computational Load Carrying between Human and Computer

The role of the operator must be determined early (or if the action team will carry all the options forward during the study) since it will impact the architecture and requirements of the system. Experience has shown that a human carrying the entire cognitive load (known as teleoperation) is well established and a computer carrying its entire cognitive load (automatic or autonomous) is realistic, but some of the mixed modes (shared control

or supervisory control) are harder to implement due to some transitioning issues (refer to T. Sheridan). A synergistic process between requirements, mode of operation, and its architecture (processor hardware and AI software) will be performed to optimize the system to be greater than the sum of its parts (as taken from the AI Technology Base). The key to the proposed AI Engineering Roadmap, besides the AI Technology Base and the Human-Machine Interface, is the architecture, both functional and physical. The architecture, which includes derived requirements in the form of AI behaviors established from AI Science, connects software to software and connects software to the hardware (including sensors, processors, power, and actuators for robotics).

The architecture develops the framework to integrate the various AI technologies into a cohesive system. Repeating, the AI Engineering goal is to develop Artificial General Intelligence (AGI) or similar. The architectures require the appropriate hooks and scars to be able to evolve the system from an initial operating configuration (IOC) to a more advanced capability over time. Not all the technologies in the AI technology base will apply to every architecture. There are subsets to AGI that can contribute to other suitable systems, e.g. autonomous systems, situational awareness systems, robot process automation (RPA) systems, etc. The ultimate artificial human is a humanoid with AGI.

The system design process is rigorous to improve our chances of success and to meet the team's objectives. Without requirements, this activity becomes a hobbyist project and there are no guarantees. The important first step to a systematic approach is to understand the problem to be solved (Ref. Polya), developing requirements that can later be verified with testing, maturing the human-computer interface to better understand the concept-of-operations, and eventually producing the architecture. This sequence is a serial, waterfall process. The architecture step signifies the beginning of the design sequence to also include analysis, modeling, and leading up to a final detail design. This work is checked by two formal design reviews: a preliminary design review (PDR) and a critical design review (CDR). This is followed by the process of hardware development (CPU, GPU, TPU, QPU) and more importantly software development (assembling AI software modules). In typical engineering designs, the assembly step is typically reserved for putting hardware together, but this time the team will focus on assembling software modules, including daily regression testing (a key function) to insure software stability. Integration involves combining the hardware with the software. Finally, testing is critical for verification and validation (V&V) of the requirements and acceptance of the deliverable AI System. There are some NSF Testbeds being developed that can be used for AI development testing and acceptance demonstrations for final customer validation (of capability). It is expected that we will have a number of smaller demonstrations along the way for management and stake holders to enhance public engagement.

Returning to the Action Plan and using the vernacular of AI language, a "plan" is a sequence of steps. The plan starts with a starting point and a goal, and similar to implementing a search tree, there are intermediate steps that can be taken to reach the goal. It is important to remember that any plan will have contingencies, which may require re-planning. This is not uncommon. The plan needs to be actionable which fits into Craik's deliberate (or sense-plan-act) paradigm in robotics. The key to any robotic system is all

about control. Thus, the OSTP AI Action Plan should be a sequence of steps that can be controlled and monitored with discrete feedback (learned from cybernetics). The first step in developing the plan is to build the **technology roadmap**. The **AI roadmaps** will form the framework for the overall technology development process. As described earlier, there are two roadmaps to be developed: an AI Science roadmap that feeds into the AI Engineering roadmap. It is important to include science, but the primary focus should be on the AI Engineering roadmap. The roadmaps need to be adaptive such that breakthroughs in science should be quickly inserted into the AI engineering roadmap.

This means that all AI activities around the world need to be captured in this roadmap, which can be difficult with individual companies developing proprietary research & development activities. The notion is not to document proprietary work, but to summarize the work in order to assess its place within the roadmap (i.e. capabilities). To stay objective, members of the roadmap team will have to sign non-disclosure statements with industry and academia. The hard part is developing the roadmap, and myself having personal experience in developing two government sponsored roadmaps (NASA and Department of Energy) and one in industry (Lockheed Martin). The Department of Energy Roadmap focused on the individual “needs” within each national laboratory in the nuclear complex and the NASA roadmap was delineated by core capabilities such as exploration and mobility for space exploration. The Lockheed Martin Roadmap was specific to Autonomy and conducted across the various Lockheed Martin business segments (delineated by domain). One key here is to get knowledgeable people to work on this effort that can work judiciously to produce the map accurately and quickly. Having worked on the National Automated Highway System Consortium (NAHSC) through the U.S. Department of Transportation, the roadmap team must be managed at a reasonable size (NAHSC had nine members) in order to make decisions in a timely manner.

The AI Roadmap is a visual way to connect strategy to actual work in order to deliver against its goals. It should be a workable plan that the country can believe in and follow. This requires a buy-in for all that are involved, in academia, industry and the government. The AI Roadmap includes goals, initiatives (both existing and new), and big themes of work that will help the country achieve both the vision and mission. The AI Roadmap is a high-level document that is a **visual representation of a strategy**; it answers the questions of what will be done, who will be involved in the work, the details of scope and resource allocation, as well as how and why certain initiatives were prioritized over others. The AI Roadmap cannot live in a silo and is essentially a communication tool for decision-makers and stakeholders. It will have the following clear depiction of dependencies, having the right amount of detail, and having an assessment of the AI technologies today (across government, industry, and academia). The AI Roadmap team needs to objectively identify strengths, weakness, and especially gaps (both domestically and internationally) in the technologies. Finally, the roadmap will be very broad with many interfaces which have complex interactions, i.e. introducing unexpected emerging properties.

Developing the Roadmap

The project starts with selecting the team to work on the project. The optimal size would be 9-12 scientist and engineers. All members should be full-time with no distractions (no part-time workers that cannot commit full-time to this endeavor). The OSTP member will not be full-time due to his/her other responsibilities. The plan is to produce a first draft of the AI Roadmap in 6-months. This is not an academic exercise. Similar to the Manhattan project or the methodology used by the Lockheed Martin “Skunkworks”, in terms of producing a new aircraft with minimal distractions, to focus on the team’s goals. Members should be co-located during the study, with maybe a free weekend every two weeks to go home to attend to personal needs. Midway (after three months), the participants are given a week holiday to recharge batteries and again rejoin family. The key is selecting workers as opposed to the opposite in industry where everyone wants to be a manager to delegate work to others. Having experienced professionals are the key.

As stated earlier, there will be two roadmaps with the primary focus on the AI Engineering roadmap. But a neuroscientist should be selected to lead the AI Science roadmap and the deliverable product would be a summary on the state-of-the art of mind research. The product of the team should be un-biased and as objective as possible. There are some interesting questions to be asked of AI Science:

- How does the mind really work?
- Can a connectionist neural network accomplish symbolic manipulation, or maybe more generically, is an integrated AI a form of neuro-symbolic AI?
- Is the modern computer system a good analogy for the mind/brain problem?
- What are the limitations of the mind and of the brain, or both?

It would be important for the entire team to be at the same place mentally, and maybe the AI Science team should be a subset of the entire AI Roadmap team. There will be instances where the AI Science team needs to make site visits (travel) to gather information from leaders in the field in order to build the AI Science roadmap.

The AI Engineering roadmap is a different beast entirely. There needs to be a state of the art of artificial intelligence technologies. For reference, there is the *Handbook of Artificial Intelligence* (Copyrighted in 1981 by editors Barr and Feigenbaum) and *Artificial Intelligence: A Modern Approach* by Stuart Russell and Peter Norvig (copyright 1995) that are both fundamental to AI teaching. There is a vast amount of background references in this subject. These two books are the most cited references on AI but does not accurately reflect the most recent stage in the development of the technology and does not incorporate the newest technologies, ideas, and features. During the development of the Department of Energy’s roadmap, we developed various working groups that visited many pertinent institutions relevant to our effort. Affectionately known as road trips, we need to continue this work to visit key institutions in this country and potentially others. This includes a combination of government, academic, and industrial affiliations. This is the reason for identifying and establishing non-disclosure agreements with the visited institutions.

Initial candidate institutions to visit include:

Academia	Private Institutions
University of California at Berkeley	Santa Fe Institute
Stanford University	Robotics and AI Institute (RAI Institute)
Massachusetts Institute of Technology	Toyota Technological Institute at Chicago
Carnegie Mellon University	Institute for Human-Machine Cognition
University of California at San Diego	
	Industry
OpenAI	Google/Alphabet AI
Tesla AI & Robotics	AI at Meta
Amazon AI	Anthropic
Apple Machine Learning Research	Vicarious
Google Deepmind	IBM Research
Nvidia	Anduril
GitHub	Microsoft AI Research

This list of institutions is preliminary and subject to change. During the information collection process, the team is required to use the Ray-Ban Meta AI glasses to document the different visits (respecting the permission to have cameras in the fact finding task). Recent technologies of interest include:

- AdaBoost
- Affinity Propagation
- Agents
- Large Language Models
- Convolutions
- Transformers
- Attention Mechanisms
- Autoencoders
- Backpropagation
- Multi-layer Perceptron
- Embeddings
- Diffusion Models
- Embeddings
- Vector Databases
- Graph Neural Networks
- Knowledge Graphs
- N-grams
- Normalization
- Random Forest
- Softmax
- Tokens
- Support Vector Machines
- Kolmogorov-Arnold Networks

- Backpropagation for Training

Again, this is only a partial list. However, recently Yann LeCun made a series of statements to abandon generative models, probabilistic models, contrastive methods, and reinforcement learning. We, as a team, must investigate where Dr. LeCun is coming from in order to understand his intentions. Having worked on a DARPA autonomous vehicle program in the 1990s, we integrated 1.6 million lines of code on five SPARC workstations to drive both on-road and offroad, resulting in a software intensive program which required arduous integration and testing. My colleagues and I served as software integrators by receiving code from many industrial and academic partners, and documenting the software was a difficult task. One of my roles on this DARPA program was to transfer technology and as a result, I had to capsulized some of the technologies demonstrated into software packages of different capabilities for transfer such as 1) teleoperation and 2) mission planning. These packages are quantified by the size of the package, understanding the required inputs and expected outputs to and from the software (described in a data flow diagrams), and all other interactions with each particular software packet. This also included hardware requirements to run the code. A key architectural issue is whether these technologies are closely coupled or loosely coupled with other technologies, e.g. data cleaning would be tightly coupled to machine learning.



The listed recent technologies of interest will not preclude classical AI technologies: search, decision trees, expert systems (another form of decision-making), reasoning, natural language processing, anomaly detection, planning (global and local), logic, symbolic manipulation, diagnostics, prognostics, inference, etc. The results of the fact-finding expeditions and the state-of-the-art for each technology will be documented in a report on the technologies investigated from the AI Technology Base. The remaining task is to document the AI Engineering roadmap. The Roadmap and the state-of-the-art report can occur in parallel.

AI Science helps to identify the individual technologies to be investigated in the AI Technology Base (can be referred to as a database, but there must be a better title for this grouping and possibly renamed). The problem statement also receives input from AI Science, and understanding the problem is a critical step. For example, a common test for individuals is to recall the problem in their own words (used on my students). So, is the problem to achieve AGI or something different? This sets the stage for the AI Engineering roadmap. At this point, the problem would be open to further debate among the team to further clarify the problem. Moving forward, proper requirements must be developed including the formalism of using “shall” and include a rationale for each requirement as justification. The requirements activity should be led by a team member with expertise in psychology (defining the behaviors required for the final product). At this time, the roadmap should focus on autonomy as the HRI, and the team can re-address the human-machine interface later. Without the human, it makes dictating the concept of operations easier. It should be remembered that all assumptions made during this

process must be validated in order to continue the plan. Otherwise, bad assumptions must be corrected and the previous analysis has to be re-done.

In a parallel thread, the team should have a notion or conceptual design of what is AI. Everyone has ideas creating an abundance of concepts to investigate. For example, typical proposals to government solicitations would start with a baseline or concept of what the offeror is planning on delivering. Some engineers would refer to this as the Initial Operating Configuration (IOC). There will be some back and forth between the requirements, architecture, concept of operations and the conceptual design in order to work out the details, and this process is to refine all the constituents, and is called synergy (term used in the NASA systems engineering manual). The functional architecture is in practice all software (what it has to do), and the physical architecture is code name for hardware. Technologies in the AI Technology base populate the functional architecture with inputs-outputs, maybe having a data dictionary, and can balance “functions” at the appropriate level of abstraction (similar to using Hatley-Pirbhai modeling, DeMarco modeling, or data flow modeling). Good examples of a high-level architecture are documented on the Vehicle Management System (on a F-35 aircraft) or the Vehicle System Manager (VSM) on the NASA Orion spacecraft. Remember, the physical architecture is designed to support the software. Typically, the architecture will take a global view of AI with the goal of looking for holes in the design. We are looking for strengths and weaknesses in the technologies. These holes are critical to the action plan for focusing this country’s needs for keeping the nation’s effort at the fore front.

Developing the architecture and populating the design from the AI Technology Base is a key step in the action plan. The architecture itself is a roadmap. The Preliminary Design Phase and the Critical Design Phase is dominated by modeling and analysis. System analysis include trade studies and a gap analysis of the critical technologies. The analysis will also include the typical analysis on computation capacity, memory, ASICs, FPGAs, power requirements, and all temporal issues from timing and update rates that impact processing cycles. For example, aircraft might incorporate ARINC 645 (makes a single processor to function like 20 smaller processors through partitioning). This technology is used on commercial aircraft today. Representation, both internal and external, is key to artificial intelligence. Some representation are AI models that include one or more of the following: mental models, world models, geometric models, script theory, frames, language models, etc. Some researchers use ensemble theory to combine multiple disparate models. Models are essential to predict the future or what will happen next, similar to the way current weather models predict tomorrow’s weather. Predictive capabilities are one of the pillars for AI, along with learning, knowledge, and natural language. Passing the critical design review delineates the go-ahead for hardware and software development. This is where the rubber-meets-the-road in AI. As the software is assembled into larger working modules, they will be integrated with its intended hardware (be it with 2 nm or 4 nm CPU technology or qubits if available). Upon working, testing is used to measure performance and verify requirements. The final test may be a demonstration of the final capability, whether AGI or other. The learning component will require a step for training the model on the data. There may be several iterations through the AI Engineering roadmap to identify different activities or to develop new capabilities.

At the high rate AI technologies are being developed today, the roadmap will have to be updated constantly to ensure currency with the state-of-the-art.

Similar to the AI Science roadmap, there are some interesting questions to be asked about the AI Engineering roadmap:

- Can we develop an AGI or superAGI system all at once or do we need to evolve the system incrementally?
- How do we integrate symbolic systems with connectionist systems (also known as sub-symbolic) ?
- Real-time performance is always desirable, but there are limitations. Are we processor limited per Moore's Law or by memory or by power limitations (or other such as requiring thermal considerations)?
- How are guiderails integrated into the system (similar to DeepSeek-V3)?
- How can we transition from specific use-cases to the general case and vice versa? Are there limitations to the level of abstractness to a level of detail required to not lose meaning?

The planned deliverables for this effort should include: the AI Science Roadmap, AI Engineering Roadmap, site visit write-ups (could be included as an appendix), physical architecture, functional architecture, test data, requirements table, performance analysis, etc. Systems engineering require complete and thorough documentation. Finally, many presentations are anticipated to be given by the team members in various forums.

Please use all or parts of this RFI to achieve your objectives. We are available for further clarification or consultation. For additional reference, please refer to:

www.routledge.com/9781032673110

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Acronyms:

AGI	Artificial General Intelligence
AI	Artificial Intelligence
ASIC	Application-specific integrated circuit
CONOPs	Concept of Operations
CPU	Central Processing Unit
CDR	Critical Design Review
FPGAs	Field Programmable Gate Arrays
GPU	Graphical Processing Unit
HCI	Human-Computer Interface
HRI	Human-Robot Interface
IOC	Initial Operating Configuration
KISS	Keep it simple, stupid
LLM	Large Language Models
Nm	Nanometer
PDR	Preliminary Design Review
RFI	Request for Information
TPU	Tensor Processing Unit
TRL	Technology Readiness Level
QPU	Quantum Processing Unit